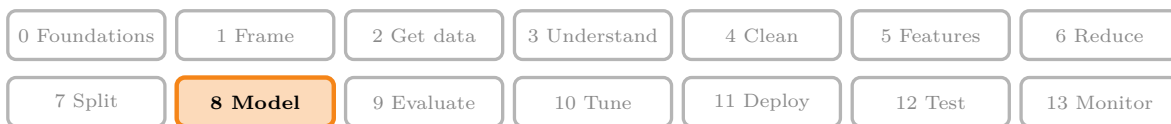


Bahdanau Attention and Luong Attention

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



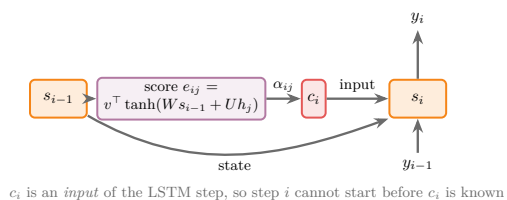
- **Builds on:** Attention mechanism ([Note 1069](#)).
- **Compare with:** Cross-attention ([Note 1082](#)).

1. Overview

Key point: Both attentions build a context vector $c_i = \sum_j \alpha_{ij} h_j$; they differ in how they score each encoder state and where the context enters the decoder. **Bahdanau** (additive) attention scores the previous decoder state s_{i-1} against h_j with a small neural network, and feeds c_i into the LSTM step. **Luong** (multiplicative) attention scores the current state s_i against h_j with a dot product, and joins c_i to the LSTM's *output*.

The [attention Note](#) introduced attention in the encoder–decoder: at every decoder step, a weighted sum of all the encoder's hidden states, with weights from a softmax over scores e_{ij} . That Note left open how the scores are computed. This Note opens the two best-known answers: Bahdanau et al. (2015) and Luong et al. (2015). Luong's dot-product score is, apart from a scaling factor, the one the transformer uses (Vaswani et al. 2017, section 3.2.1).

(a) Bahdanau: attention *before* the LSTM step



(b) Luong: attention *after* the LSTM step

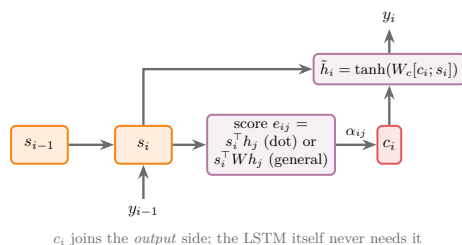


Figure 1: Where attention sits in the two designs. (a) Bahdanau: the previous state s_{i-1} is scored, and c_i is an input of the LSTM step. (b) Luong: the LSTM step runs first, its new state s_i is scored, and c_i is combined with s_i before the output layer

2. Prerequisites

- [Attention Note](#): context vectors, alignment scores, softmax over the input positions.
- [Encoder–decoder Note](#): encoder, decoder, teacher forcing.
- [Dot product and cosine similarity Note](#): the dot product as a measure of similarity.
- [Forward propagation Note](#): a dense layer as a matrix product.

3. Recap: what both must compute

Key point: Scores e_{ij} , weights α_{ij} from a softmax of the scores over j , context $c_i = \sum_j \alpha_{ij} h_j$. Only the score function and the wiring differ.

For “turn off the lights” $\rightarrow$ “light band karo”, decoder step 1 needs $c_1 = \alpha_{11}h_1 + \alpha_{12}h_2 + \alpha_{13}h_3 + \alpha_{14}h_4$, step 2 needs c_2 with new weights, and so on: (number of input words) $\times$ (number of output words) weights in all. Each weight is a word-to-word similarity: α_{11} says how much “turn” counts when writing “light”. The question of this Note is how to get the raw scores e_{ij} .

4. Bahdanau attention

Key point: A feed-forward network with one hidden layer scores the pair (s_{i-1}, h_j) : $e_{ij} = v^\top \tanh(W[s_{i-1}; h_j])$. The context vector is then an input of decoder step i .

4.1 What the score depends on

Key point: On the encoder state h_j and on the decoder’s *previous* state s_{i-1} , which holds what has been translated so far.

As the [attention Note](#) (section 6) explains, α_{ij} must depend on h_j , the input word being judged, and on what the decoder has already written, which is stored in s_{i-1} . To compute α_{11} , the weight of “turn” for the first output word, we need h_1 and s_0 ; for α_{21} we need h_1 and s_1 . Bahdanau et al. (2015) do not choose a formula for the score; they let a small feed-forward network learn it, the **alignment model**.

4.2 The alignment network, step by step

Key point: Concatenate s_{i-1} with every h_j , pass the rows through a hidden layer with tanh, then through one output unit: one score per input word. Softmax gives the weights.

Take hidden states of size 4 and an alignment network with 3 hidden units and 1 output unit. At decoder step 1:

1. **Concatenate.** Join s_0 (4 numbers) with each of $h_1, \dots, h_4$ (4 numbers each). Stacked, the four rows form a 4×8 matrix X , one row per input word.
2. **Hidden layer.** Multiply by the 8×3 weight matrix W and apply tanh: a 4×3 matrix. A bias can be added before the tanh.

3. **Output unit.** Multiply by the 3×1 vector v : 4 numbers, the scores $e_{11}, \dots, e_{14}$.
4. **Softmax** over the 4 scores gives $\alpha_{11}, \dots, \alpha_{14}$, and the weighted sum gives c_1 .
5. **Decoder step.** c_1 , s_0 and the input y_0 (`<start>`) go into the LSTM, which outputs the first word, “light”, and the new state s_1 .

At step 2 the same network runs again with s_1 in place of s_0 . In the 4×8 matrix only the left half changes; the encoder states on the right stay the same. The changing decoder state is why every step gets different weights. The network’s weights are shared across all decoder steps, like a time-distributed dense layer, and are updated by backpropagation together with the two LSTMs.

1. **In words:** concatenate the two states, apply one tanh hidden layer, then one output unit; softmax over the input positions.
2. **Formula** (Bahdanau et al. 2015, appendix A.1.2):

$$e_{ij} = v^\top \tanh(W[s_{i-1}; h_j]) = v^\top \tanh(W_a s_{i-1} + U_a h_j), \quad \alpha_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})}, \quad c_i = \sum_j \alpha_{ij} h_j$$

The two forms are the same: multiplying the joined vector $[s; h]$ by W equals multiplying s by the left half of W and h by the right half, and adding.

3. **Example:** the decoder state $s = [0.5, -0.2, 0.8, 0.1]$ and four encoder states, the first being $h_1 = [0.3, -0.5, -0.9, -1.0]$ (the Notebook lists all four and the weights W and $v = [0.1, -0.4, 0.2]$). For $j = 1$, the joined row is $[0.5, -0.2, 0.8, 0.1, 0.3, -0.5, -0.9, -1.0]$. Times W it gives $[-0.06, 0.47, 0.44]$; tanh gives $[-0.060, 0.438, 0.414]$; times v :

$$e_1 = 0.1(-0.060) - 0.4(0.438) + 0.2(0.414) = -0.099$$

The four scores are $(-0.099, -0.011, 0.449, -0.116)$, and softmax turns them into the weights $(0.208, 0.227, 0.360, 0.205)$. The score function has $8 \times 3 + 3 = 27$ learned numbers.

Because the score adds a term from s_{i-1} to a term from h_j inside the tanh, Bahdanau attention is also called **additive attention** (Vaswani et al. 2017, section 3.2.1). Since $U_a h_j$ does not depend on i , it can be computed once per sentence (Bahdanau et al. 2015, appendix A.1.2).

5. Luong attention

Key point: Two changes. The score uses the *current* decoder state s_i , and it is a simple product: $e_{ij} = s_i^\top h_j$ (dot) or $s_i^\top W_a h_j$ (general). The context vector is joined to s_i after the LSTM step, not fed into it.

Luong, Pham and Manning (2015) kept the goal and changed the means.

5.1 A simpler score

Key point: Two similar vectors have a large dot product. Using the dot product as the score needs no network at all.

The aim of the score is not to approximate some exact function; it is to find which encoder states are useful now. A similarity measure does that job, and the simplest one is the dot product (the [dot product Note](#)): large when two vectors point the same way, small or negative when they do not. Luong et al. (2015, section 3.1) proposed three content-based scores:

Name	Score e_{ij}	Learned parameters
dot	$s_i^\top h_j$	none
general	$s_i^\top W_a h_j$	one matrix W_a
concat	$v_a^\top \tanh(W_a[s_i; h_j])$	W_a and v_a , as in Bahdanau

The dot score requires s_i and h_j to have the same size; the general score lifts that requirement and lets the model learn which directions of similarity matter (SLP3 §14.8). Because the score multiplies the two states, this family is called **multiplicative attention** (Vaswani et al. 2017, section 3.2.1).

1. **In words:** multiply the decoder state and each encoder state element by element and add up (dot), or first transform the encoder state by a learned matrix (general).
2. **Formula:**

$$e_{ij} = s_i^\top h_j \text{ (dot),} \quad e_{ij} = s_i^\top W_a h_j \text{ (general)}$$

3. **Example:** the same $s = [0.5, -0.2, 0.8, 0.1]$ and $h_1 = [0.3, -0.5, -0.9, -1.0]$:

$$e_1 = 0.5(0.3) + (-0.2)(-0.5) + 0.8(-0.9) + 0.1(-1.0) = 0.15 + 0.10 - 0.72 - 0.10 = -0.57$$

The four dot scores are $(-0.57, 0.35, 0.25, 0.87)$, with weights $(0.100, 0.251, 0.227, 0.422)$: no learned numbers at all. With a 4×4 matrix W_a (16 learned numbers, in the Notebook) the general scores are $(0.477, -0.254, 0.975, -0.887)$, with weights $(0.296, 0.142, 0.486, 0.076)$. The three functions rank the four states differently. Here the states and the weights are random numbers; in a trained model, training sets them so that the useful encoder states get the high scores, whichever function is used.

5.2 The current state, and the context on the output side

Key point: The LSTM step runs first. Its new state s_i is scored against the encoder states, and $\tilde{h}_i = \tanh(W_c[c_i; s_i])$ feeds the softmax.

Bahdanau's decoder goes $s_{i-1} \rightarrow \alpha_i \rightarrow c_i \rightarrow s_i$: the context must be ready before the LSTM step. Luong's goes $s_i \rightarrow \alpha_i \rightarrow c_i \rightarrow \tilde{h}_i$ (Luong et al. 2015, section 3.1):

1. The decoder LSTM takes y_{i-1} and s_{i-1} and produces s_i , with no context vector.
2. s_i is scored against every h_j ; softmax gives α_{ij} ; the weighted sum gives c_i .

3. c_i and s_i are joined and passed through a dense layer with tanh: the **attentional hidden state** $\tilde{h}_i = \tanh(W_c[c_i; s_i])$.
4. A softmax layer on $\tilde{h}_i$ gives the next word: $p(y_i) = \text{softmax}(W_s \tilde{h}_i)$.
5. s_i (not $\tilde{h}_i$) and the word y_i go on to step $i + 1$.

Scoring with s_i uses the most recent state, which already includes the latest word written. Figure 1 shows the two wirings side by side.

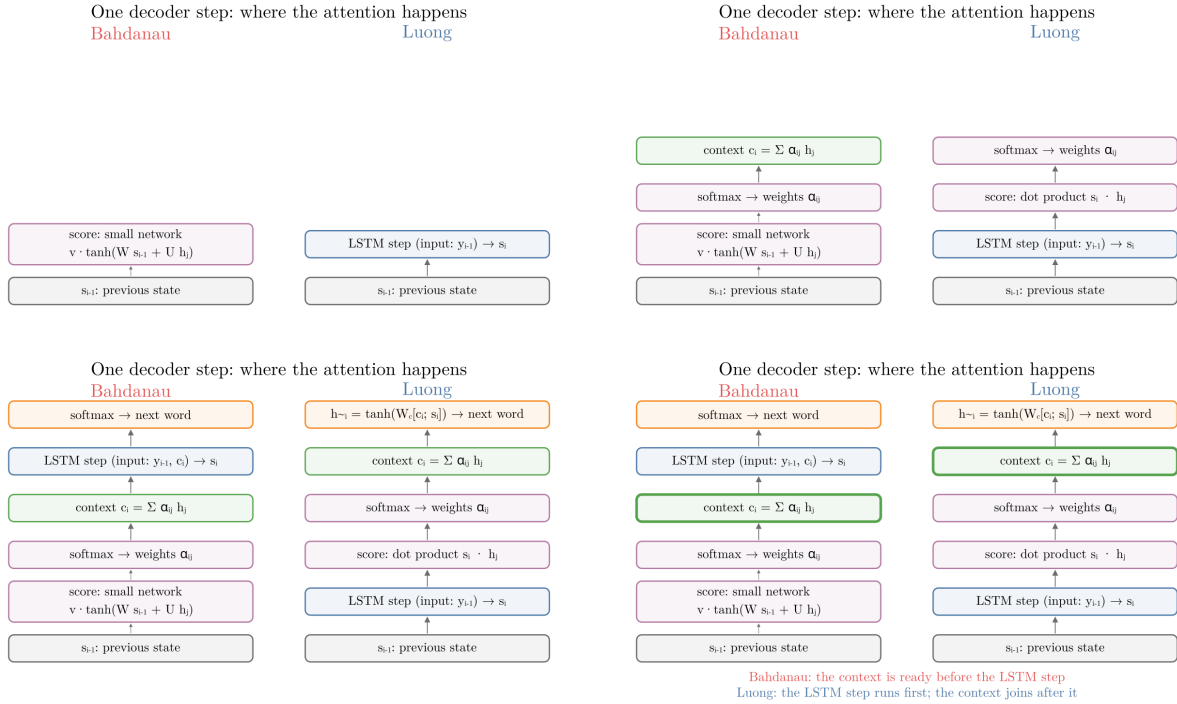


Figure 2: One decoder step in the two designs, built up one operation at a time from the bottom. Bahdanau scores the previous state and computes the context before the LSTM step; Luong runs the LSTM step first, scores the new state by a dot product, and joins the context after the step

Figure 2 builds the same step in both designs, one operation at a time, from the bottom up. Watch where the green context box lands: below the LSTM step on the left, above it on the right.

Python: Luong attention for all decoder steps at once (training, teacher forcing). S : decoder states, shape (batch, m , 256); hs : encoder states, (batch, n , 256).

```

keys = Wa(hs) if general else hs # general:  $s^T W$ 
 $\hookrightarrow h = s \cdot (W h)$ 
e = tf.einsum("bid,bjd->bij", S, keys) # scores, (batch,
 $\hookrightarrow m, n$ )
a = tf.nn.softmax(e - 1e9 * pad[:, None, :], axis=-1) # padding gets
 $\hookrightarrow$  weight 0

```

```
ctx = tf.einsum("bij,bjd->bid", a, hs) # context
↪ vectors, (batch, m, 256)
logits = out(Wc(tf.concat([ctx, S], -1))) # Wc: Dense(256,
↪ "tanh")
```

Extra: Luong et al. (2015, section 3.3) also propose **input feeding**: $\tilde{h}_{i-1}$ is joined to the next input, so that the model remembers its past alignment choices. With input feeding the LSTM again needs the previous step’s attention before it can run. The Notebook leaves it out, as in Luong’s basic global model of section 3.1.

6. The two compared

Key point: On the same data, Luong’s dot attention was the best of the three and the fastest to train: test BLEU 31.6 against 25.1 for Bahdanau’s additive attention, with a third fewer parameters and less than half the time per epoch. Luong’s general score did worst (17.7): its model learned to put all its weight on the last word.

6.1 Summary of the differences

Key point: Which decoder state, which score, where the context goes.

	Bahdanau (2015)	Luong (2015)
Decoder state scored	previous, s_{i-1}	current, s_i
Score e_{ij}	$v^\top \tanh(W_a s_{i-1} + U_a h_j)$	$s_i^\top h_j$ (dot) or $s_i^\top W_a h_j$ (general)
Other name	additive attention	multiplicative attention
Learned parameters in the score	W_a, U_a, v	none (dot) or W_a (general)
Where c_i goes	input of the LSTM step	joined to s_i after the step: $\tilde{h}_i = \tanh(W_c[c_i; s_i])$

Luong et al. (2015, section 3.1) describe their path $s_i \rightarrow \alpha_i \rightarrow c_i \rightarrow \tilde{h}_i$ as “simpler” than Bahdanau’s. Vaswani et al. (2017, section 3.2.1) add that the two are similar in theoretical complexity, but dot-product attention “is much faster and more space-efficient in practice, since it can be implemented using highly optimized matrix multiplication code”.

6.2 The experiment

Key point: Three attention models on the same 60,000 sentence pairs, 3 runs each.

The Notebook trains the attention model of the [attention Note](#) (Bahdanau) and two Luong models (dot and general) on the same English–French data, with the same bidirectional

encoder and the same decoder LSTM of 256 units, for 12 epochs, 3 runs each. The Luong models follow equations 5–7 of Luong et al. (2015) without input feeding.

	Bahdanau (concat)	Luong dot	Luong general
Parameters	8.42 million	5.66 million	5.73 million
Seconds per epoch (median, shared GPU)	51	22	20
Final training loss	0.77	0.80	1.06
Test BLEU, all 1,000 sentences	25.1	31.6	17.7
Test BLEU, sentences of 11–16 words	20.0	26.8	12.0

The Bahdanau numbers differ slightly from those of the [attention Note](#) (25.7) because GPU training is not exactly repeatable. The differences between the models are much larger than that.

Parameters. Bahdanau’s model has more because its output layer reads $[s_i; c_i]$ (512 numbers) and its LSTM reads $[y_{i-1}; c_i]$; Luong’s output layer reads the 256-number $\tilde{h}_i$. The scores themselves are a small part: the dot score has no parameters, and general adds one 256×256 matrix (65,536 numbers).

Time. In Bahdanau’s model, c_i is an input of LSTM step i , so each step must wait for the attention of that step: the decoder runs as a loop of small operations. In Luong’s model the LSTM never needs c_i , so during training (with teacher forcing) the decoder LSTM runs over the whole gold sentence in one call, and the attention for all steps at once is two matrix products, $SH^\top$ for the scores and AH for the contexts. Each epoch took less than half the time.

Quality. Luong’s dot attention gave the best translations at all lengths. The general score did worst; Luong et al. (2015, Table 4) also found dot better than general for global attention (BLEU 18.6 against 17.3 on English–German).

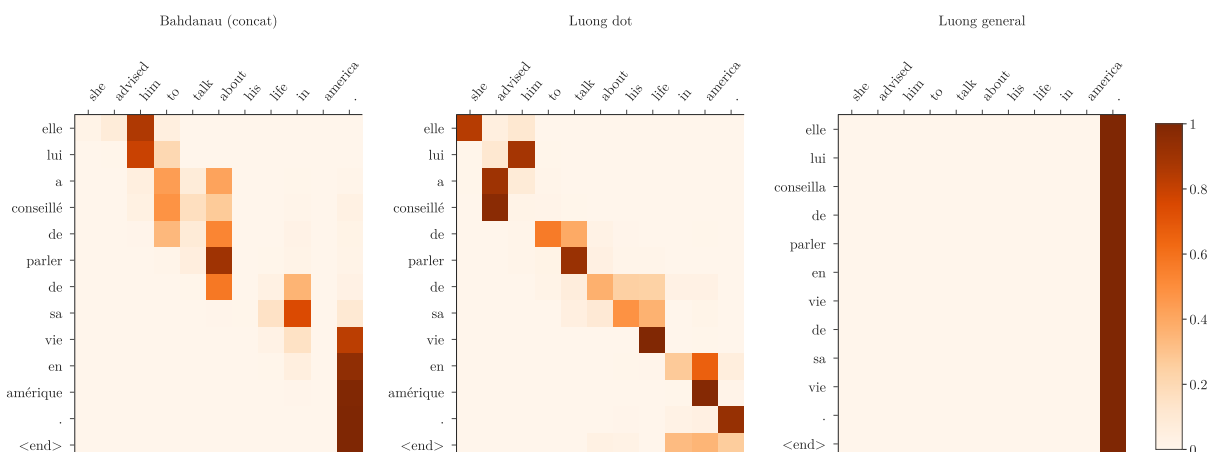


Figure 3: Attention weights of the three trained models on the same real test sentence. Bahdanau and Luong dot both translate it exactly; Luong general puts all its weight on the final full stop

Figure 2 shows why the general model did badly. On the test sentence “she advised him to talk about his life in america .”, the dot model’s weights form a clean diagonal: “elle” looks at “she”, “lui” at “him”, “conseillé” at “advised”, “parler” at “talk”, “vie” at “life”, “Amérique” at “america”. Bahdanau’s model also forms a band, but it often sits to the right of the matching word (“parler” looks at “about”); with the same encoder, data and plotting code the dot model does not shift, so the shift is no plotting offset. Attention weights need not match a word alignment, and the same off-by-one attention has been reported in a trained Bahdanau-type system ([attention Note](#), section 8). The general model puts a weight of 1 on the final “.” at every step. Its context vector is then the same at every step: the state of the last word, whose forward half has read the whole sentence. The model has turned itself back into a plain encoder–decoder with one fixed summary, and its translation of this sentence goes wrong at the end (“elle lui conseilla de parler en vie de sa vie .”).

Extra: Luong et al. (2015, Table 1) compared their full systems with earlier attention systems on English–German: with unknown-word replacement, their single model with global attention and the dot score reached 20.5 BLEU (Table 4) and their best single model, with local attention, 20.9, against 19.0 for an RNNsearch (Bahdanau-style) system of Jean et al. (2015) with the same replacement. Their systems differ in more than the attention (stacked LSTMs, reversed input, dropout), so the gap is not due to the score function alone.

7. Summary

	Bahdanau	Luong
Score	additive: small network on $[s_{i-1}; h_j]$	multiplicative: $s_i^\top h_j$ or $s_i^\top W_a h_j$
Decoder state used	previous (s_{i-1})	current (s_i)
Context vector enters	the LSTM step	the output layer, via $\tilde{h}_i$
In the Notebook	BLEU 25.1, 51 s per epoch	dot: BLEU 31.6, 22 s per epoch

- Both compute weights α_{ij} by a softmax over scores and a context vector $c_i = \sum_j \alpha_{ij} h_j$.
- Bahdanau scores the previous decoder state with a one-hidden-layer network; the context is an input of the next LSTM step.
- Luong scores the current decoder state with a dot product (or a bilinear form), and joins the context to the state after the LSTM step.
- The dot product needs no parameters and, without input feeding, lets attention for all decoder steps be computed with matrix products.
- The dot-product score is the one the transformer’s self-attention builds on.

8. Sources

Built from

- CampusX, “Bahdanau Attention Vs Luong Attention”, YouTube, <https://www.youtube.com/watch?v=>

Other references

- Bahdanau, D., Cho, K. and Bengio, Y. (2015). Neural Machine Translation by Jointly Learning to Align and Translate. *ICLR 2015*. arXiv:1409.0473. Section 3.1 (eq. 4–6); appendix A.1.2 (the alignment model $v_a^\top \tanh(W_a s_{i-1} + U_a h_j)$, pre-computing $U_a h_j$).
- Luong, M.-T., Pham, H. and Manning, C. D. (2015). Effective Approaches to Attention-based Neural Machine Translation. *EMNLP 2015*. arXiv:1508.04025. Section 3 (eq. 5–8: attentional hidden state, the dot, general and concat scores); 3.1 (comparison with Bahdanau et al.); 3.3 (input feeding); Table 1; Table 4 and section 5.3 (choice of score function).
- Vaswani, A. et al. (2017). Attention Is All You Need. *NeurIPS 2017*. arXiv:1706.03762. Section 3.2.1 (additive and dot-product (multiplicative) attention).
- Jurafsky, D. and Martin, J. H. *Speech and Language Processing*, 3rd ed. draft (19 August 2026), web.stanford.edu/~jurafsky/slp3. Chapter 14, §14.8 (dot-product attention; the bilinear score allows states of different sizes). Cited as SLP3.
- Tatoeba project and manythings.org/anki: the English–French sentence pairs, distributed as `fra-eng.zip` with the Keras examples.

9. Key terms

Term	Meaning
Bahdanau attention (additive)	Attention whose score is a one-hidden-layer network on the previous decoder state and an encoder state: $v^\top \tanh(W_a s_{i-1} + U_a h_j)$
Luong attention (multiplicative)	Attention whose score multiplies the current decoder state with an encoder state: $s_i^\top h_j$ or $s_i^\top W_a h_j$
Score function	The function that gives the raw score e_{ij} of encoder state j for decoder step i
Dot score	$s_i^\top h_j$; needs no parameters and states of equal size
General score	$s_i^\top W_a h_j$, with a learned matrix W_a
Concat score	$v_a^\top \tanh(W_a [s_i; h_j])$, the additive form
Attentional hidden state $\tilde{h}_i$	$\tanh(W_c [c_i; s_i])$: the decoder state combined with the context vector, read by Luong's output layer
Input feeding	Luong's option of feeding $\tilde{h}_{i-1}$ into the next decoder step